

Dependent Variable (Target Variable)

- Nutritional Status

Independent Variable(Feature Categories)

1. Demographic Factors -
 - Age (v012) - Represents the respondent's age in years.
 - Age Group (v013) - Categorized age ranges.
 - Marital Status (v502) - Indicates whether the respondent is married or not.
2. Socioeconomic Factors -
 - Education level (v106) - Indicates highest level of education.
 - Years of Education (v107) - Total years spent in education.
 - Wealth Index (v190) - Composite measure of household wealth.
3. Occupational Factors -
 - Working Status (v714) - Indicates whether the respondent is employed.
4. Household factors -
 - Household Size (v136) - Number of people in the household.
 - Water Source (v114) - Type of drinking water.
 - Sanitation Facility (v116) - Type of toilet facility.
5. Geographic factors -
 - Region (v024) - Geographic region of residence.
 - Urban/Rural (v025) - Type of residence.
6. Health and Maternal factors -
 - Healthcare Access (v467d) - Difficulty in accessing health facilities.
 - Total Children Ever Born (v201) - Indicator of reproductive burden.
 - Antenatal Visits (m14) - Number of prenatal care visits.
 - Pregnancy status

SUPERVISED LEARNING:

1. Random Forest Classifier - Handles both categorical and numerical data and works with large datasets.
2. XGBoost - High accuracy , handle missing values well.
3. Logistic Regression - Simple and interpretable.

4. Decision Tree - Easy to understand.
5. Naive Bayes - Fast , works well with categorical data.

UNSUPERVISED LEARNING:

1. K-Means Clustering

age group

- 1: "15-19",
- 2: "20-24",
- 3: "25-29",
- 4: "30-34",
- 5: "35-39",
- 6: "40-44",
- 7: "45-49"

Marital Status

- 1 → Married
- 0 → Never married

Education level

- 0 → No education
- 1 → Primary education
- 2 → Secondary education
- 3 → Higher education

Wealth index

- 1 → Poorest
- 2 → Poorer
- 3 → Middle
- 4 → Richer
- 5 → Richest

Working status

- 0 → Not working
- 1 → Working

OCCUPATION_TYPE

- 0 → Not working
- 1 → Professional/Technical
- 3 → Sales
- 4 → Agriculture
- 5 → Skilled manual
- 6 → Unskilled manual
- 7 → Household work
- 9 → Other
- 98 → Missing / Don't know

Water_source

- 11 → Piped water (into dwelling/yard)
- 12 → Public tap / standpipe
- 14 → Tube well / borehole
- 31 → Surface water (river, stream, pond)
- 42 → Unprotected well
- 43 → Unprotected spring

Sanitation

- 11 → Flush toilet (to sewer/septic tank)
- 12 → Flush toilet (pit latrine)
- 13 → Ventilated Improved Pit (VIP) latrine
- 44 → No facility / open defecation

Region

- 1 → Jammu & Kashmir
- 2 → Himachal Pradesh
- 3 → Punjab
- 4 → Chandigarh
- 5 → Uttarakhand
- 6 → Haryana
- 7 → Delhi
- 8 → Rajasthan
- 9 → Uttar Pradesh
- 10 → Bihar
- 11 → Sikkim
- 12 → Arunachal Pradesh
- 13 → Nagaland
- 14 → Manipur
- 15 → Mizoram
- 16 → Tripura
- 17 → Meghalaya
- 18 → Assam
- 19 → West Bengal
- 20 → Jharkhand
- 21 → Odisha
- 22 → Chhattisgarh
- 23 → Madhya Pradesh
- 24 → Gujarat
- 25 → Daman & Diu
- 26 → Dadra & Nagar Haveli

- 27 → Maharashtra
- 28 → Andhra Pradesh
- 29 → Karnataka
- 30 → Goa
- 31 → Lakshadweep
- 32 → Kerala
- 33 → Tamil Nadu
- 34 → Puducherry
- 35 → Andaman & Nicobar Islands
- 36 → Telangana
- 37 → Ladakh

Urban_rural

- 1 → Urban
- 2 → Rural

Healthcare_access

- 0 → Not a problem
- 1 → A big problem
- 2 → Don't know / Missing

Pregnancy_status

- 0 → Not pregnant
- 1 → Pregnant